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REVIEW ARTICLE

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Prevalence of the use of chemical restraint in the management of challenging behaviours associated with adult mental health conditions: A meta-synthesis

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Accessible summary

What is known on the subject?

- The use of chemical restraint in emergency situations is to control aggression or violence to protect consumers and staff and is to be used as a last resort when all other behavioural control alternatives have been exhausted.
- Chemical restraint involves the use of medication.

What the paper adds to existing knowledge?

- This is the first paper to systematically review the prevalence of the use of chemical restraint.
- The use of chemical restraint is common across emergency departments and acute psychiatric inpatient units worldwide.
- Consumers who are restrained are also likely to receive chemical restraint.
- The studies reviewed demonstrate that there is a need for more standardized data collection so that clearer comparisons can be made between healthcare settings and countries.

What are the implications for practice?

- Implications of practice include the need to report adverse events post-chemical restraint as these can deleteriously affect a person's physical health.
- Identification of and staff education about first-line management techniques before chemical restraint is administered are significant to reduce its use.

Abstract

Introduction: Chemical restraint is used to manage uncontrolled aggression, agitation or violent behaviours of consumers with mental health disorders admitted to acute psychiatric or emergency settings.

Aim: This systematic review aimed to synthesize the international prevalence of chemical restraint for non-consenting adults.

Method: PsycINFO, CINAHL, MEDLINE/PubMed and Google Scholar databases were searched for peer-reviewed literature published between January 1996 and July 2018. This paper reports on data extracted from retrospective audits of chemical restraint practice.

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Results: Forty-eight papers were included. The median prevalence of use of any restraint was 21.2% (25th% 8.0 to 75th% 36.3). Median prevalence of people who were chemically restrained, of all people restrained in any manner, was 43.1% (25th% 22.9% to 75th% 70.7%). Of all people admitted to facility(ies), the median prevalence of chemical restraint was 7.4% (25th% 2.7 to 75th% 17.6). There was no statistically significant difference in any prevalence measure considering healthcare setting or country. There was a significant decrease over the review period in the use of any restraint, including chemical restraint.

Discussion and implications for practice: This is the first known comprehensive meta-view of chemical restraint use worldwide, highlighting the need for standardized data collection to enable comparisons between healthcare settings and countries.

Relevance statement: There is an international imperative to reduce or eliminate the use of coercive practices, such as physical and chemical restraint, in mental health care. This study provides important information for mental health nursing by synthesizing the international prevalence of chemical restraint for non-consenting adults. This meta-view of the worldwide use of chemical restraint can inform ongoing efforts to reduce its use.

KEYWORDS

chemical restraint, prevalence, restraint

1 | INTRODUCTION

Chemical restraint, also known as rapid tranquilization, is the use of prescribed drugs for sedation in situations where people with mental health disorders present with severe agitation, aggression or violent behaviours that pose a danger to themselves or others (Knott, Bennett, Rawet, & Taylor, 2005). It is a last resort option in situations where harm cannot otherwise be averted (Hochstrasser et al., 2018). Chemical restraint most often occurs in pre-hospital situations, hospital emergency departments (EDs) or other acute psychiatric settings such as psychiatric intensive care units or acute care wards (Isbister, 2017; Taylor et al., 2016; Wilhelm, Schacht, & Wagner, 2008; Zun, 2018; Zun, Wilson, & Nordstrom, 2017).

While chemical restraint is most often used to manage violence and aggression (Hu, Muir-Cochrane, Oster, & Gerace, 2019), recent literature suggests an increasing prevalence of agitation in consumers presenting to healthcare settings and the increasing pressure placed on healthcare providers to maintain safety (Foo, Duffull, Calver, Schneider, & Isbister, 2016; Isbister, 2017; Weiland, Ivory, & Hutton, 2017). Agitation can occur from medical conditions or substance use (Preval, Klotz, Southard, & Francis, 2005). It can also occur as a result of primary psychiatric illness (Mantovani et al., 2013).

The debate about the value, place and application of chemical restraint and other coercive measures is long-standing (Deitch et al., 2014). Providing a least restrictive environment in which to deliver care has been a long aspiration in mental health settings and is recognized as an indicator of quality care. Over 20 years ago, Hiday, Swartz, Swanson, and Wagner (1997) reported on the ongoing mental health law debate over the justification of civil commitment of those affected by mental illness. This included the tension between civil libertarians' concerns with deprivations to free choice and movement, and mental health practitioners' concerns about managing risk in situations where consumers or others could be harmed. As part of this, there has been international agreement on the need to reduce and, where possible, eliminate the use of coercive measures (seclusion, physical and chemical restraint; Goulet, Larue, & Dumais, 2017; LeBel et al., 2014). However, the safety of health professionals and people receiving care is paramount and chemical restraint is an accepted though controversial practice to be used as a last resort when all other options to de-escalate the situation have been explored.

There have been some attempts to collect information on prevalence of chemical restraint across countries, for instance the EUNOMIA Project that collected prevalence data from 10 Western countries, demonstrating that chemical restraint use in psychiatric inpatient facilities ranged from 81% of involuntarily admitted patients in Sweden to 31% in Greece (Raboch et al., 2010). The authors note these variations might relate to differences in clinical traditions and societal attitudes. Previous systematic reviews have explored the prevalence of physical restraint and seclusion (Beghi, Peroni, Gabola, Rossetti, & Cornaggia, 2013; De Hert, Dirix, Demunter, & Correll, 2011), but there is no current, comprehensive research into the prevalence of incidents requiring the use of chemical restraint. This systematic review aimed to synthesize the international prevalence of chemical restraint use for non-consenting people with mental health disorders to control aggression, severe agitation or violent behaviours from a comprehensive systematic

review of 22 years of literature on the use of chemical restraint. The study question was: What is the international prevalence of chemical restraint, delivered by any means, to manage non-consenting adults with severe agitation, aggression and/or violent behaviours?

2 | METHODS

This paper reports on a subset of papers from what we are referring to as "the full data set," a comprehensive systematic review of primary research literature of any hierarchy on the use of chemical restraint to manage non-consenting adults with mental health disorders exhibiting severe agitation, aggression and/or violent behaviours (see Muir-Cochrane, Oster, & Grimmer, 2019). The systematic review was registered with PROSPERO (protocol number CRD42017055258) on 13 January 2017. The section below summarizes the review processes for the full data set, followed by a description of the subset of data for the present review of the prevalence of chemical restraint.

2.1 | The full data set

Databases comprised PsycINFO, CINAHL, MEDLINE/PubMed and Google Scholar. Relevant search terms were discussed with the university liaison Librarian, who then independently conducted the search. Broad search terms included chemical restraint, forced medication, coerced medication, intramuscular medication, rapid tranquilization, restraint, mental health, agitation, aggression, violence, psychiatric, ED and variants/other terms identified in a preliminary review of the literature. For the full search strategy, see Muir-Cochrane et al. (2019). The search sought literature published since January 1996 to ensure that changes over time in contemporary mental health practices could be examined. The search start date reflected findings in a literature review of research into chemical restraint in acute care from 1980 to 2008 (Jarrett, Bowers, & Simpson, 2008) where only four of 14 papers were published pre-1996.

A systematic, five-phase search was applied to ensure that all relevant literature was identified. The first search was of library databases from January 1996 to January 2017. The reference lists of the potentially relevant articles identified in this search were then hand-searched for other potentially relevant literature. To address questions raised within the team about eligibility and inclusion criteria, a second search was undertaken of the same databases from January 1996 to April 2018, using expanded search terms relating to the location of chemical restraint occurrence (e.g., ambulance, paramedic, ED). Duplicate papers with search one were excluded, and the reference lists of any new papers identified in the second search were hand-searched. In the final, fifth step, the primary literature contained within the included systematic reviews was scrutinized, and new studies not identified in the previous four searches were sourced.

Titles and abstracts of potentially relevant literature were screened by two independent reviewers, and those articles that potentially addressed the research questions were read in full. Peer-reviewed journal articles reporting any study design were eligible for inclusion if the literature was available in English and full text, and if it reported on the use of chemical restraint for non-consenting adults (aged 18–65) with acute agitation, aggression, and/or violent behaviours. Research conducted in any healthcare setting (acute hospital care, psychiatric or general hospital intensive care, EDs) was considered. All included papers needed to specifically indicate that consumers did not consent to administration of chemical restraint (i.e., that it was coercive, and delivered under duress or by force). The inclusion of only those articles reporting on the use of chemical restraint for non-consenting adults was because some articles also included information on people who had consented to medication, indicating treatment rather than chemical restraint (Muir-Cochrane et al., 2019).

Studies were excluded if they were published prior to January 1996, included children and adolescents <18 years or people with non-psychiatric conditions such as dementia or neurological diseases; or were unpublished conference abstracts. We also excluded non-peer-reviewed literature and papers unavailable in English. Literature was excluded if subjects were reported to have consented prior to being administered pharmacological methods to calm agitation or aggression, or if research was undertaken in prison settings.

2.2 | Subset of data for this review

Prevalence is an expression of the proportion of people with a specific issue in a population. It can be expressed as point prevalence (measured at a specific point in time); period prevalence (measured over a particular time period); or lifetime prevalence (event ever occurred during a person's lifetime; National Institute of Mental Health, 2017). While point prevalence is generally derived from cross-sectional studies, period and life prevalence can be determined retrospectively from existing data sources (such as audits, retrospective surveys, or record reviews) or prospectively (where events of interest are flagged as they occur). This study assessed period prevalence of chemical restraint from retrospectively collected data identified from the full systematic review data set (NHMRC III-3 studies; Merlin, Weston, & Tooher, 2009). Retrospective data (commonly audits or record reviews) were sought for this analysis because it provided "closed" data sets derived from similar sources, with similar purposes, clear time frames and similar approaches to data collection. Similar approaches have been taken in systematic reviews of self-reported child abuse (Bonoldi et al., 2013) and chronic pain (Mansfield, Sim, Jordan, & Jordan, 2016). We were mindful, however, that the retrospective nature of record reviews and audits may incur a high likelihood of similar biases (such as failure to identify all relevant records, data missing from identified records, and differences in decisions to apply chemical restraint of different healthcare providers in the one setting).

The intention was to undertake a meta-synthesis if homogenous data from two or more studies were available. A meta-synthesis is a method of combining quantitative data that come from different sources (Urquhart, 2010). This review was underpinned by a realist synthesis framework to focus the study inclusion criteria, data extraction and data standardization approaches to the "what, when and how" of chemical restraint. Realist synthesis assists in analysing data in systematic literature reviews where there are complex interventions (such as chemical restraint; Rycroft-Malone et al., 2012). Using this framework, each potentially relevant article was assessed for the reasons given for chemical restraint administration and the ways in which chemical restraint events were described. The framework also assisted in standardizing the different ways in which prevalence data were reported across studies (Pawson, Greenhalgh, Harvey, & Walshe, 2005; Rycroft-Malone et al., 2012). To be eligible for inclusion in the meta-synthesis, studies needed, at a minimum, to clearly state that chemical restraint had been provided (using a definition, policy or explanation that indicated chemical restraint had been delivered without consumer consent) and to report information on the time period of data collection, the aim of the study, the country and healthcare setting, the total number of people admitted to the healthcare setting in that time period, the total number of consumers for whom any restraint had been applied in that time period, and the number of people for whom chemical restraint had been delivered.

As this review was a synthesis of retrospective audit data, where no quality evaluation was undertaken, not all reporting requirements for systematic reviews were likely to be met. Study processes and findings were reported as best as possible according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses framework (Moher, Liberati, Tetzlaff, & Altman, 2009), which has been utilized previously by systematic reviews of coercive practices (Goulet et al., 2017).

2.3 | Synthesis of findings

Data were extracted (where possible) on country of research; year(s) in which research occurred; study purpose; definition of chemical restraint; time frame of data collection (calculated in months); setting; the level of data reported (consumer, incident); overall number of people admitted (or episodes) to the setting; the total number (episodes) subjected to restraint of some form; the number (episodes) subjected to chemical restraint; demographic information on both/ either group (age, gender); adjunctive non-chemical restraint methods (if reported); methods of administration; dosages and timing of intervention delivery (if reported); and adverse events (if reported).

Study methodological quality was appraised using the Joanna Briggs Institute Critical Appraisal Checklist for reporting prevalence data (Munn, Moola, Lisy, Riitano, & Tufanaru, 2015). The checklist contains nine items for assessing risk of bias in the studies, assessed by answering yes (Y), no (No) or unsure (U) against each item. The "Yes" answers were aggregated to calculate compliance (total of nine JBI criteria calculated as a percentage of possible total).

Where chemical restraint prevalence was reported in a paper as only a rate, and the total number of consumers in the sample was also reported, the raw number of people receiving chemical restraint was calculated, if possible. Raw numbers were required for meta-synthesis. Similarly, raw numbers for gender were calculated if papers only reported percentages. Where age was reported in age categories (e.g., 18–30, 31–40 years), mean age was estimated by summing the number of people in each age category and multiplying this by the median age in each category, and then summing this information across the age categories and dividing the total estimated age number by the sample.

2.4 | Analysis

This paper considers only the consumer-level data, as this was most commonly reported. Incidence data were not considered in this meta-analysis. Data on the number of consumers admitted to the facility(ies), the total number of restrained consumers (physically or chemically) and the numbers of consumers who were chemical restrained were standardized to a one-month time period. Prevalence (expressed as percentage) was calculated on the consumer-level data, for three comparisons: (any) restraint administered/total consumers/month; chemical restraint administered/total admissions/month; and chemical restraint administered/total consumers restrained/month. For comparisons between countries, the per-country papers were organized alphabetically, and when there were multiple papers from one country, papers were organized chronologically. The data were graphed for visual inspection. For comparisons between healthcare settings, data were organized into ED (general and psychiatric) and hospital wards (including psychiatric wards, general wards and psychiatric intensive care wards). ANOVA models were used to test differences in prevalence between countries and settings. Considering the 22 years of chemical restraint literature, the hypothesis was tested that chemical restraint prevalence would decrease over time, in line with international concerns regarding its ethics and safety (Deitch et al., 2014; Goulet et al., 2017; LeBel et al., 2014). To facilitate this investigation, data were arranged chronologically, and scatter plots were constructed. Trendlines and regression statistics were then superimposed to calculate change over time in prevalence estimate. R^2 statistics (p values) were calculated for each trendline. MS Excel (Windows Corporation 2017) was used for these calculations. Gender differences in samples were calculated using MedCalc Statistical Software version 19.1 (MedCalc Software bvba, Ostend, Belgium). Forest and funnel plots (and associated statistics) were also calculated using MedCalc version 19.1. The decision to apply random- or fixed-effects models was made on the size of the Q and I^2 statistics.

3 | RESULTS

From the complete systematic review data set of 311 articles, there were 69 potentially relevant papers reporting on retrospectively collected data (NHMRC Level III-3). After excluding papers that did not address the aims of this paper, or which did not provide sufficiently clear or extractable details on chemical restraint (*N* = 17), and the four papers reporting the same findings generated from the same data sets (duplicate publications), there were 48 relevant studies.

Not every study reported all requisite data for all prevalence calculations (total admissions, total given restraint of some type, total chemically restrained). Forty papers reported consumer-level data, and eight reported incident-level data. Thirty-one papers reported complete consumer-level data, six reported complete incident-level data, and one paper reported mixed consumer-level and incident-level data (38 studies in total). However, the 10 studies that did not report complete minimum data to calculate prevalence were retained in the review data set, as they provided other information such as demographics or types of chemical restraints used. The study inclusion flow chart is shown in Figure 1. The 48 included studies are reported in Table 1, with data comprising principal author, year of publication, country(ies) of research, setting of research, time period over which data had been collected, data level (consumer, episode) and study aim. The excluded studies and studies that reported on the same data sets already reported in included studies are listed in Table 2.

Approximately 40% papers reported on multi-site data collection, either within a number of facilities in a specified region, or within facilities in a country or in specific types of facilities across countries. Consequently, there were large differences in sample size, particularly compared with papers where only one facility was investigated. Where multi-sites provided audit data, it was rarely made clear why those sites had been chosen, or how generalizable the findings were. The most common justification provided for auditing sites was to test changes in use of chemical restraint as a result of policy, education or workforce interventions.

While papers were only included from 1996 onwards, the data reported in some had been collected earlier (Hyde, Harrower-Wilson, & Morris, 1998 [1992–1994]; Kaltiala-Heino, Korkeila, Tuihimaki, Tuori, & Lehtinen, 2000 [1996]; Knutzen, Sandvik, Hauff, Opjordsmoen, & Friis, 2007 [1996–1997]; Ohlenschlaeger & Nordentoft, 2008 [1999–2001]; Segal, Bola, & Watson, 1996 [1981–1986]; Tuohimaki et al., 2003 [1996]; Wynn, 2002 [1989–1994]). The oldest data set was reported by Segal et al. (1996 [1981–1986]). The papers reporting on consumer-level data reported audits taken from 0.5 to 120 months (median 12 months [25th%–75th% 9–27 months]), while the papers reporting on incident-level data reported on data sets collated from 2 to 84 months (median 21.5 months [25th%-75th% 10.5–34.5 months). There was a median time delay of 4 years (25th% to 75th% 2–6 years) between completing data collection and publishing. The largest delay was 10 years (Knutzen et al., 2007).

3.1 | Definition of chemical restraint

Eighteen papers provided a definition of chemical restraint that underpinned delivery of care for consumers with mental health disorders and/or had been used to identify records for auditing purposes. Definitions provided in these papers are outlined in Table 3, along with principal author, year and country information.

While the definitions of chemical restraint differed in these 18 papers, they consistently described the use of medications, delivered by force and/or without consumers' consent, to control agitated or violent behaviours associated with mental health disorders that potentially endangered the consumer and or/others. All but two of these papers further state that chemical restraint was not treatment; rather, it was an emergency management strategy to control a potentially dangerous situation. Treatment options were determined later, once calm had been restored (Bowers et al., 2017; Knutzen et al., 2013, 2014). The reason for applying chemical restraint was reported variably in the included papers, including emergency calls for assistance (e.g., Code Black calls), consumers breaching the rules of the psychiatric setting or challenging behaviours to self, staff or other consumers.

Thus, to be included in this review, all included papers needed to specifically indicate that consumers did not consent to administration of medication to manage their behaviours. All included papers provided detail on the nature of non-consent and how decisions were made to apply chemical constraint; for instance, consumers' relatives provided proxy consent, doctors took responsibility for administering restraint as per hospital policies (such as emergency management or consumer aggression episodes) or legislation (e.g., mental health acts).

3.2 | Methodological quality

Table 4 reports compliance per study for the individual JBI critical appraisal criteria (Y = compliance; N = non-compliance; U = uncertain).

Methodological quality in the included studies varied between 66.7% and 100% compliance with the methodological quality criteria. Thirty of the 48 included papers (62.5% total included papers) scored well (8/9 or 9/9 of the JBI critical appraisal criteria). Of these papers, one (Staggs, 2015) scored 100%, and the remaining 29 papers scored 8/9 (88.9%). The remaining 18 papers scored moderately on the JBI criteria, with 13 papers (27.1% included papers) scoring 7/9 criteria (78.8% total JBI criteria), and the remaining five papers (10.4% included papers) scored poorest (6/9 (66.7%) JBI criteria).

The sample was adequately described in all papers because of the use of protocols and clear descriptions of the reasons for delivering chemical restraint. Thirty-nine papers (81.2% included papers) reported recruiting records of all consumers who had presented to the facility(ies) during the specified time period. Six papers used consecutive recruitment (meaning that there were generally no data provided on the volume of consumers to the setting), three papers used random sampling of records for audit purposes, and the remaining two papers used convenience sampling. The criteria where there was consistently poor

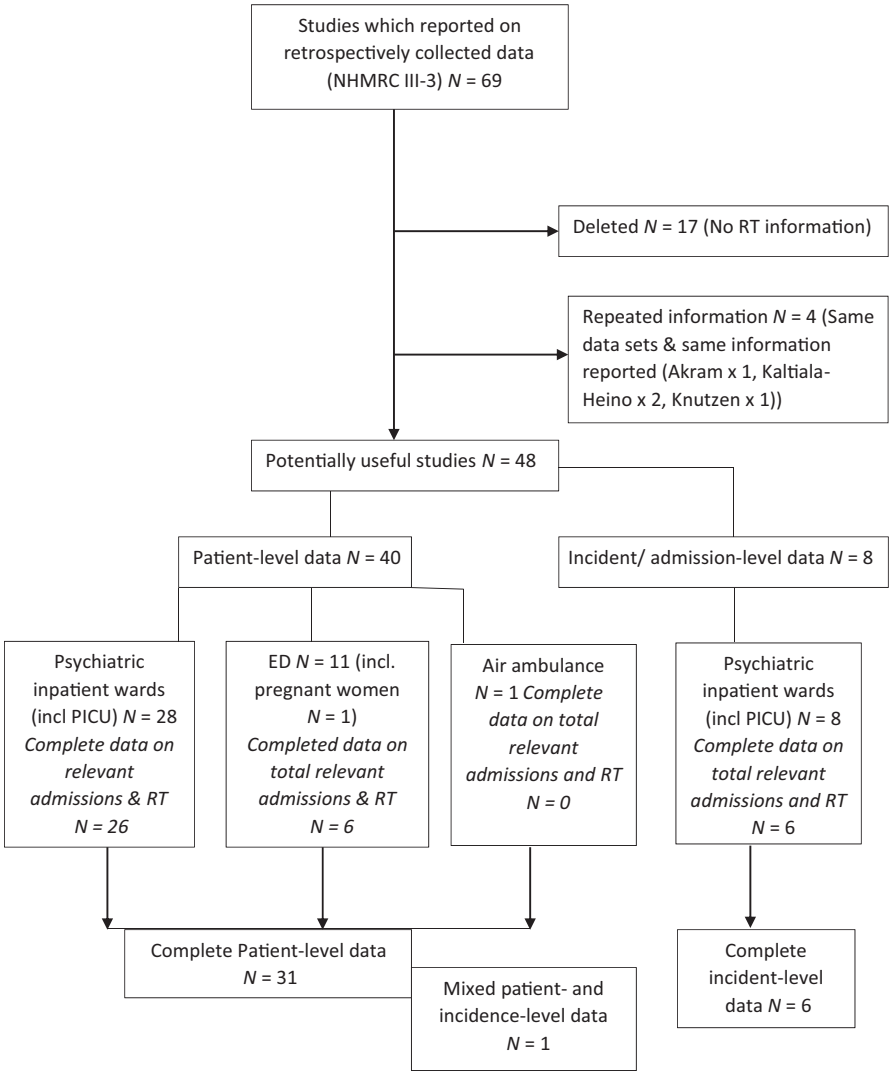

FIGURE 1 Article inclusion flow chart (RT = restraint)

compliance were formal evidence of sample size calculation relevant to JBI criterion 3. All but one paper reported a proxy measure for sample size calculation as the time frame of data capture and the expected number of consumers to attend the facility in that time as the rationale.

3.3 | Prevalence of chemical restraint

This paper reports on consumer-level data, from 32 papers that reported on 302,602 consumers who presented to the facilities over a total of 779 months. Of these, 49,765 had been restrained chemically (which may have included physical restraint) to manage agitation and/or violent aggressive behaviours, with 27,546 being chemically restrained. Combining data from all papers prior to time period standardization, 16.4% consumers were restrained physically or chemically, and of these, just over half (55.3%) were chemically restrained. Chemical restraint was reported for 9.1% overall of all consumers in the data set. For comparison, data on the number of consumers admitted to the facility(ies), the total number of restrained consumers (physically or chemically) and the numbers of consumers who were chemical restrained were standardized to a one-month time period.

After each study data set had been standardized to a common per-month time frame, the median prevalence (25th% to 75th%) of people who were restrained in any manner, of all presenting consumers, was 21.2% (8.0% to 36.3%). The country-by-country prevalence figures are reported in Figure 2, sorted largest to smallest. Median prevalence (25th%–75th%) of people who were chemically restrained, of all consumers who were restrained in any manner, was 43.1% (22.9%–70.7%). Figure 3 reports these country-by-country prevalence figures (sorted largest to smallest). Median prevalence (25th%–75th%) of people who were chemically restrained of all presenting consumers was 7.4% (2.7%–17.6%). Figure 4 reports these prevalence figures country by country.

There was no statistically significant difference in any prevalence measure considering emergency and ward settings, or between countries (p = .24). There was a significant decrease, however, over the chronological publication period (2006–2018) in reporting the use of restraint in general, and chemical restraint in particular (p = .01). This decrease was identified by the negative slope and R^2 value of the trendline. Considering all restrained consumers as a percentage of all admitted consumers, there was an overall 7.8% decrease over the review period (see Figure 5, R^2

TABLE 1 Summary of included studies

Author	Year	Country	Setting	Time period (months)	Reporting level	Aims
Akram et al.	2014	Scotland	PICU	0.5	Consumers	Investigate psychotropic PRN meds
Andersen et al.	2016	Denmark	Psych ward	24	Consumers	Describe rates of coercion and restraint
Beysard et al.	2017	Swiss	ED	23	Consumers	Describe characteristics of mechanically restrained consumers
Bilanakis et al.	2011	Greece	Psych ward	36	Consumers	Report on chemical restraint practices
Bowers et al.	2005	UK, Italy, Greece	Psych ward	8	Consumers	Compare rates of disturbed behaviours and containment methods
Bowers et al.	2013	UK	Psych ward	9	Consumers	Identify common pathways of conflict and containment from event to event
Bowers et al.	2017	UK	Psych ward	60	Consumers	Compare across units how high-risk behaviours are managed
Chan and Chung	2005	HK	ED	36	Consumers	Study the practice of seclusion and factors associated with it
Choudhury et al.	2011	UK	Psych ward	4	Consumers	Scrutinize clinical practice to determine compliance with NHS restraint policy
Downes et al.	2009	Australia	ED	12	Consumers	Describe characteristics of consumers with acute behavioural disorders and their restraint
Flammer et al.	2013	Germany	Psych ward	12	Consumers	Explore impact of consumers' characteristics on risk of coercion
Flammer and Steinert	2015	Germany	Psych ward	12	Consumers	Assess frequency of use of involuntary medication
Hopper et al.	2015	USA	ED	93	Consumers	Examine efficacy and safety of ketamine for acute anxiety
Hyde et al.	1998	UK	PICU	26	Consumers	Audit violent incidents and restraint
Kaltiala-Heino et al.	2000	Finland	Psych ward	6	Consumers	Examine frequency and type of restraint use, and types of consumers
Kazemi-Bajestani et al.	2011	Netherlands, Iran, USA	Psych ED	12	Consumers	Assess correlation between psychiatric ED visits and lunar cycles
Knight	2005	USA	Psych ward	12	Consumers	Examine clinical symptoms and behaviours at point of admission of people who experienced seclusion and/or restraint
Knott et al.	2007	Australia	ED	6	Consumers	Determine nature of mental health presentations
Knutzen et al.	2007	Norway	Psych ward	24	Consumers	Examine restraint rate, demographics, type
Knutzen et al.	2013	Norway	Psych ward	24	Consumers	Describe frequencies of restraint types
Ladavac et al.	2007	USA	Psych ED	24	Pregnant women	Describe how pregnant women with aggressive incidents are managed
Lay et al.	2011	Switzerland	Psych ward	12	Consumers	Describe coercive measures and role of predictive factors
Le Cong et al.	2012	Australia	Air ambulance	40	Consumers	Describe use of ketamine during air ambulance retrieval
Leerbeck et al.	2017	Denmark	Psych ward	24	Consumers	Describe use of coercion by time of day and week
Lehane and Rees	1996	Wales	PICU	3.5	Consumers	Examine response to incidents that would normally have led to seclusion
Macdonald et al.	2012	USA	ED	26	Consumers	Assess relative safety of intramuscular (IM) haloperidol and IM olanzapine

(Continues)

TABLE 1 (Continued)

Author	Year	Country	Setting	Time period (months)	Reporting level	Aims
Mah et al.	2015	Canada	Psych ward	48	Consumers	Describe rates of restraint use
Mann-Poll et al.	2018	Holland	Psych ward	120	Consumers	Describe change in rates of chemical restraint over changed management phases
Martel et al.	2016	USA	ED	6	Consumers	Review use and safety of IV olanzapine
Nakamura et al.	2013	Japan	Psych ward	36	Consumers	Review use of chemical and mechanical restraint
Norum et al.	2013	Norway	Psych ward	24	Consumers	Explore voluntary and compulsory admissions to psychiatric hospitals
Øhlenschlæger and Nordentoft	2008_2	Denmark	Psych ward	12	Consumers	Investigate use of coercive measures with consumers with schizophrenia
Raboch et al.	2010	10 EU countries (EUNOMIA project)	Psych ward	30	Consumers	Assess and compare use of coercive measures
Reitan et al.	2018	Norway	Psych ward	96	Consumers	Describe characteristics of use of pharmacological and mechanical restraint
Robling et al.	1997	UK	Psych ward	9	Consumers	Review practice of management of acute behavioural disturbance
Ryan and Bowers	2006	UK	Psych ward	12	Consumers	Describe nature, frequency and outcome of restraint-related events
Schillerstrom et al.	2004	USA	Psych ED	1	Consumers	Compare blood profiles between IM-medicated and non-IM-medicated psychiatric consumers
Segal et al.	1996	USA	Psych ED	60	Consumers	Investigated differential restraint of white and African American consumers
Shahpesandy et al.	2015	UK	PICU	12	Consumers	Identify pharmacological management of acutely disturbed behaviour in consumers requiring restraint
Shepherd et al.	2014	UK	Psych ward	19	Consumers	Establish patterns of practice in restraint
Staggs	2015	USA	Psych ward	84	Consumers	Assess trends in use of seclusion and restraint
Stewart et al.	2012	UK	Psych ward	12	Consumers	Describe events leading to special observation
Talukdar et al.	2016	UK	Psych ward	12	Consumers	Assess whether physical health monitoring of consumers receiving IM medications was per NICE guidelines
Tuddenham and Logan	2005	UK	Psych ward	6	Consumers	Describe medication used to manage aggressive episodes
Tuohimaki et al.	2003	Finland	Psych ward	6	Consumers	Describe the extent the danger-to-others criterion is used for involuntarily hospitalized consumers
Wheeler et al.	2007	NZ	PICU	2	Admissions	Evaluate the impact of CPG for medication management of behavioural disturbance
Woo and Ma	2007	USA	Psych ward	24	Admissions	Investigate differences in resource use, quality improvement, outcomes
Wynn	2002	Norway	Psych ward	66	Episodes of aggression	Examine staff strategies for applying emergency procedures for violent or aggressive consumers

TABLE 2 Excluded studies and studies reporting data sets that were duplicates of included studies

Author	Year	Country	Setting	Time period (months)	Reporting level
Chase	2002	USA	ED	12	
Freeman	2009	USA	psych ward	36	Consumers
Gudjonsson	2004	UK	psych ward	36	Episodes of aggression
Hillam	1996	UK			
Innes	2012	UK	psych ward		
Innes	2013	UK	psych ward		
Jacob	2016	USA	psych ward	72	Episodes of aggression
Knutzen	2011	Norway	psych ward	24	Consumers
Lepping	2006	Wales, Ireland, Germany, Netherlands	12	patients	
Loynes	2012	UK			
Morgan	2007	Australia			
Srebnik	2007	USA			
Steinert	2007	Germany	psych ward	12	Cases
Steinert	2014	Germany			
Stein-Parbury	2008	Australia			
Stewart	2012a	UK	psych ward		
Stewart	2012c	UK	psych ward		
Stolker	2005	Netherlands	psych ward	24	Consumers
Sullivan	2005	USA	psych ward	36	Events/1000 pt_days
Vruwink	2012	Netherlands	psych ward		
Wilson	2012	USA			
Wynn	1996	Norway	psych ward	66	Episodes of aggression
Duplicate reported data sets (additional papers provided no new information)					
Akram	2014_1	Scotland			
Kaltiala-Heino	2003 (same data as 2,000 paper)	Finland	Psych ward		
Kaltiala-Heino	2003_1	Finland			
Knutzen	2013	Norway			

value of the superimposed trendline .078, p = .02). Considering chemically restrained consumers as a percentage of all restrained consumers, there was an overall 15.5% decrease over time (see Figure 6, R^2 value of the superimposed trendline .155, p < .05), and considering chemically restrained consumers as a percentage of all admitted consumers, there was an overall 22.6% decrease over time (R^2 value of the superimposed trendline .226, p = .009; see Figure 7).

TABLE 3 Definitions of chemical restraint reported in the included studies

Akram et al.	2014	Scotland	Medicines given for short-term management of disturbed/ violent behaviour including PRN meds taken from an agreed RT protocol
Andersen et al.	2016	Denmark	Either voluntary or involuntary coercion (belt, physical, forced meds) to treat psychosis, severe agitation, life-threatening illness
Bilanakis et al.	2011	Greece	Medicines for management of behaviour in consumers who are disturbed
Bowers et al.	2005	UK (Italy, Greece)	To manage high levels of uncontrolled aggression as scored on Patient Conflict Checklist
Bowers et al.	2013	UK	To manage high levels of uncontrolled aggression as scored on Patient Conflict Checklist
Bowers et al.	2017	UK	To manage high levels of uncontrolled aggression as scored on Patient Conflict Checklist
Chan and Chung	2005	HK	To manage violence or threat of violence
Choudhury et al.	2011	UK	Use of medicines to deal with acute behavioural disturbance not manageable by other means
Downes et al.	2009	Australia	Management if acute behavioural emergencies requiring Code Black activation
Flammer et al.	2013	Germany	Application of medicines by force
Flammer and Steinert	2015	Germany	Any medication (oral or IM) delivered against a consumer's will
Knutzen et al.	2013	Norway	Restraint is a method used to prevent injuries to consumers or others regardless of the consumer's legal status
Mah et al.	2015	Canada	Medicines administered to achieve an immediate level of control over agitation and threatening, destructive or assaultive behaviours to prevent harm to self or others
Raboch et al.	2010	EUNOMIA project	Forced medication refers to activities using restraint or strong psychological pressure (involving at least three staff members) to administer medication against the consumer's will
Shepherd et al.	2014	UK	Restraint aims to quickly calm severely agitated consumers in order to reduce risk of imminent and serious violence to self and others
Staggs	2015	USA	Administration of medicines (oral, IM, IV) by physical means (restraint)
Talukdar et al.	2016	UK	Intramuscular (IM) medicines used for restraint to manage violent/ aggressive behaviour
Wynn	2002	Norway	Use of restraint to manage violent behaviours

3.4 | Associations between consumer factors and chemical restraint

Twenty-four papers provided data on age and gender of the overall audit sample (total N = 80,112). There was a significantly larger percentage of men (51.3% (95% CI 50.9–51.6), p = .03) compared with 49.7% women. Median age was similar (men 38.3 years (25th%–75th% 34.3–42 years); women 38.6 years (25th%–75th% 35–41.3 years), p = .27). Ten papers provided demographic information on gender and age of consumers receiving chemical restraint. There was again a significantly larger percentage of men (53.2% [95% CI 52.6–53.8]) compared with 46.8% women (p = .02).

Twenty-four papers provided details of diagnosis of people receiving chemical restraint (N = 39,866). Some papers reported this information as ICD-10 codes, others by text descriptors. Papers consistently noted that unless the consumer was already known to them, diagnosis was made after chemical restraint had been applied. Schizophrenia (including schizoid features) was most commonly reported (41.3%), followed by "other" diagnoses (which often seemed to combine other psychiatric disorders) (28.9%). Mood disturbances accounted for 10.9%, followed by self-harm (5.9%), personality disorder (4.6%), drug and alcohol effects (3.6%), organic presentations (brain trauma, etc.) (3.3%) and acute psychiatric disorders (undifferentiated) (1.4%).

TABLE 4 Critical appraisal scores (JBI checklist for prevalence studies)

Author (date)	JBI Item 1	JBI Item 2	JBI Item 3	JBI Item 4	JBI Item 5	JBI Item 6	JBI Item 7	JBI Item 8	JBI Item 9
Akram, Slavin, and Davies (2014)	Y	Y	U	Y	Y	Y	Y	Y	Y
Andersen and Nielsen (2016)	Y	Y	U	Y	Y	Y	Y	Y	Y
Beysard, Yersin, and Carron (2017)	Y	Y	U	Y	Y	Y	Y	Y	Y
Bilanakis, Papamichael, and Peritogiannis (2011)	U	Y	U	Y	Y	Y	Y	N	Y
Bowers et al. (2005)	Y	Y	U	Y	Y	Y	Y	Y	Y
Bowers et al. (2013)	Y	Y	U	Y	Y	Y	Y	Y	Y
Bowers et al. (2017)	Y	Y	U	Y	Y	Y	Y	Y	Y
Chan and Chung (2005)	Y	Y	U	Y	Y	Y	Y	U	Y
Choudhury, Dewsbery, Williams, and Hovey (2011)	U	Y	U	Y	Y	Y	Y	N	Y
Downes, Healy, Page, Bryant, and Isbister (2009)	U	Y	U	Y	Y	Y	Y	Y	Y
Flammer, Steindert, Eisele, Bergk, and Uhlmann (2013)	Y	Y	U	Y	Y	Y	Y	Y	Y
Flammer and Steinert (2015)	Y	Y	U	Y	Y	Y	Y	N	Y
Hopper et al. (2015)	U	Y	U	Y	Y	Y	Y	Y	Y
Hyde et al. (1998)	Y	Y	U	Y	Y	Y	Y	N	Y
Kaltiala-Heino et al. (2000)	Y	Y	U	Y	Y	Y	Y	Y	Y
Kazemi-Bajestani, Amirsadri, Samari, and Javanbakht (2011)	Y	Y	U	Y	Y	Y	Y	Y	Y
Knight (2005)	Y	Y	U	Y	Y	Y	Y	Y	Y
Knott, Pleban, Taylor, and Castle (2007)	Y	Y	U	Y	Y	Y	Y	Y	Y
Knutzen et al. (2007)	Y	Y	U	Y	Y	Y	Y	Y	Y
Knutzen et al. (2013)	Y	Y	U	Y	Y	Y	Y	Y	Y
Ladavac, Dubin, Ning, and Stuckeman (2007)	Y	Y	U	Y	Y	Y	Y	N	Y
Lay, Nordt, and Rossler (2011)	Y	Y	U	Y	Y	Y	Y	Y	Y
Le Cong, Gynther, Hunter, and Schuller (2012)	Y	Y	U	Y	Y	Y	Y	N	Y
Leerbeck, Mainz, and Boggild (2017)	Y	Y	U	Y	Y	Y	Y	Y	Y
Lehane and Rees (1996)	U	Y	U	Y	Y	Y	Y	N	Y
Macdonald et al. (2012)	Y	Y	U	Y	Y	Y	Y	Y	Y

(Continues)

TABLE 4 (Continued)

Author (date)	JBI Item 1	JBI Item 2	JBI Item 3	JBI Item 4	JBI Item 5	JBI Item 6	JBI Item 7	JBI Item 8	JBI Item 9
Mah, Hirdes, Heckman, and Stolee (2015)	Y	Y	U	Y	Y	Y	Y	Y	Y
Mann-Poll et al. (2018)	Y	Y	U	Y	Y	Y	Y	Y	Y
Martel, Klein, Rivard, and Cole (2016)	U	Y	U	Y	Y	Y	Y	Y	Y
Nakamura et al. (2013)	Y	Y	U	Y	Y	Y	Y	Y	Y
Norum, Olsen, Nybrodahl, and Sorgaard (2013)	Y	Y	U	Y	Y	Y	Y	Y	Y
Ohlenschlaeger and Nordentoft (2008)	Y	Y	U	Y	Y	Y	Y	Y	Y
Raboch et al. (2010)	Y	Y	U	Y	Y	Y	Y	Y	Y
Reitan, Helvik, and Iversen (2018)	Y	Y	U	Y	Y	Y	Y	Y	Y
Robling, Flynn, Hunt, Gregory, and McBreen (1997)	N	Y	U	Y	Y	Y	Y	U	Y
Ryan and Bowers (2006)	N	Y	U	Y	Y	Y	Y	Y	Y
Schillerstrom, Schillerstrom, and Taylor (2004)	U	Y	U	Y	Y	Y	Y	Y	Y
Segal et al. (1996)	U	Y	U	Y	Y	Y	Y	Y	Y
Shahpesandy et al. (2015)	U	Y	U	Y	Y	Y	Y	U	Y
Shepherd, Parker, and Arif (2014)	Y	Y	U	Y	Y	Y	Y	Y	Y
Staggs (2015)	Y	Y	Y	Y	Y	Y	Y	Y	Y
Stewart, Bowers, and Ross (2012)	Y	Y	U	Y	Y	Y	Y	Y	Y
Talukdar, Ludlam, Pout, and Lekka (2016)	N	Y	U	Y	Y	Y	Y	U	Y
Tuddenham and Logan (2005)	Y	Y	U	Y	Y	Y	Y	Y	Y
Tuohimaki et al. (2003)	Y	Y	U	Y	Y	Y	Y	Y	Y
Wheeler et al. (2007)	Y	Y	U	Y	Y	Y	Y	Y	Y
Woo and Ma (2007)	Y	Y	U	Y	Y	Y	Y	Y	Y
Wynn (2002)	Y	Y	U	Y	Y	Y	Y	Y	Y

Item 1. Was the sample frame appropriate to address the target population?; Item 2. Were study participants sampled in an appropriate way?; Item 3. Was the sample size adequate?; Item 4. Were the study subjects and the setting described in detail?; Item 5. Was the data analysis conducted with sufficient coverage of the identified sample?; Item 6. Were valid methods used for the identification of the condition?; Item 7. Was the condition measured in a standard reliable way?; Item 8. Was there appropriate statistical analysis?; Item 9. Was the response rate adequate, and if not, was the low response managed appropriately?

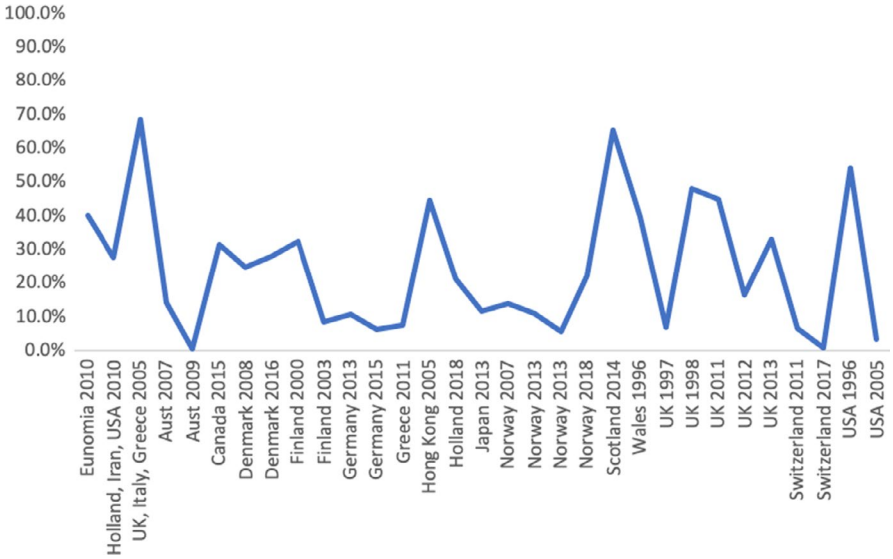

FIGURE 2 One-month prevalence of any physical or chemical restraint used in the sample of presenting consumers/country

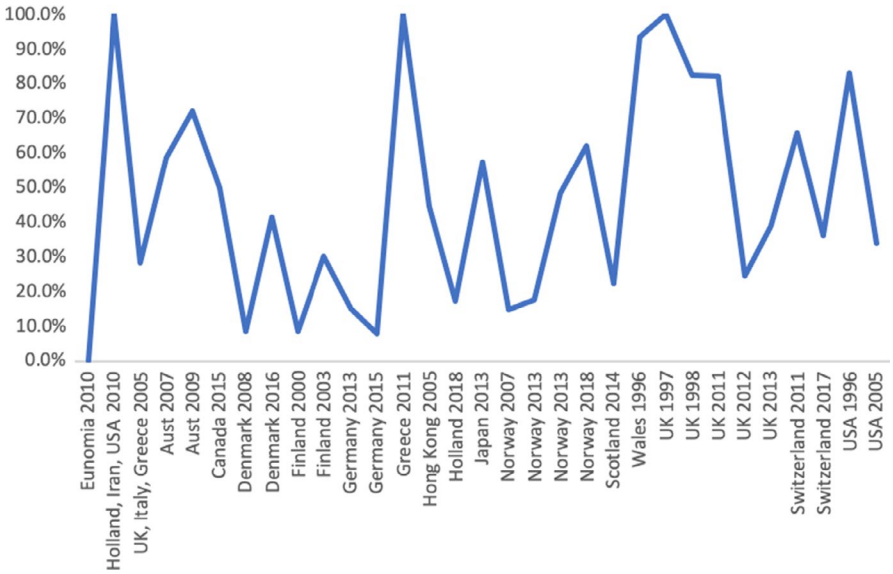

FIGURE 3 One-month prevalence of chemical restraint, used as part of any form of restraint used in consumers requiring behaviour management/country

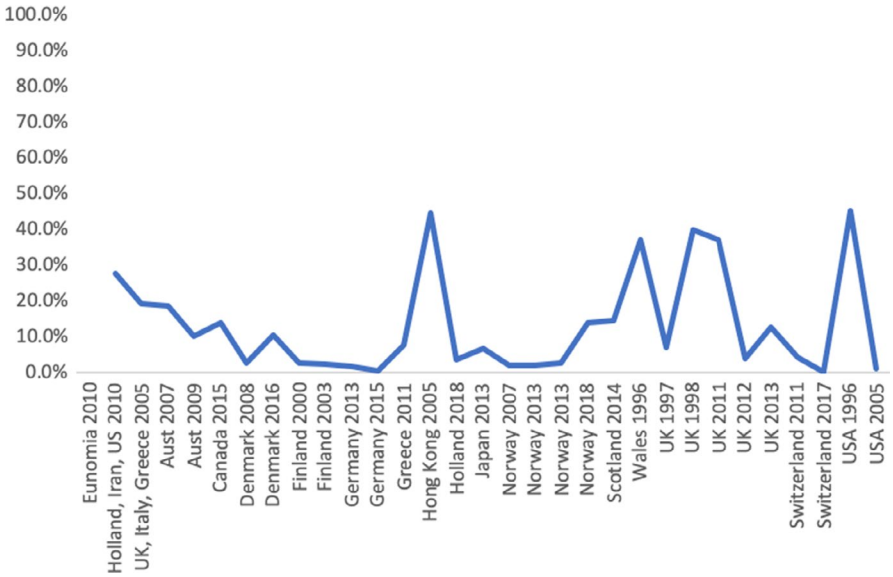

FIGURE 4 One-month prevalence of chemical restraint, as a percentage of all admitted consumers/country

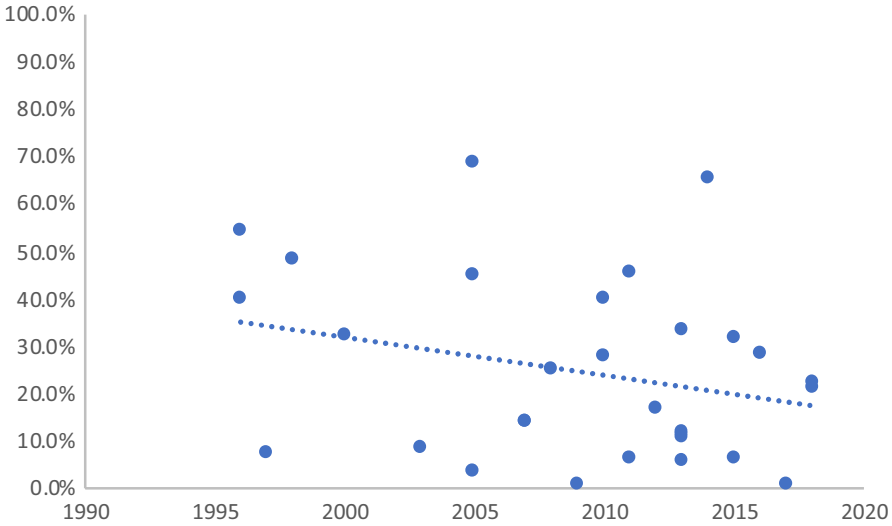

FIGURE 5 Chronological trends in use of any restraint (as a percentage of all admitted consumers/month) in year order of publication

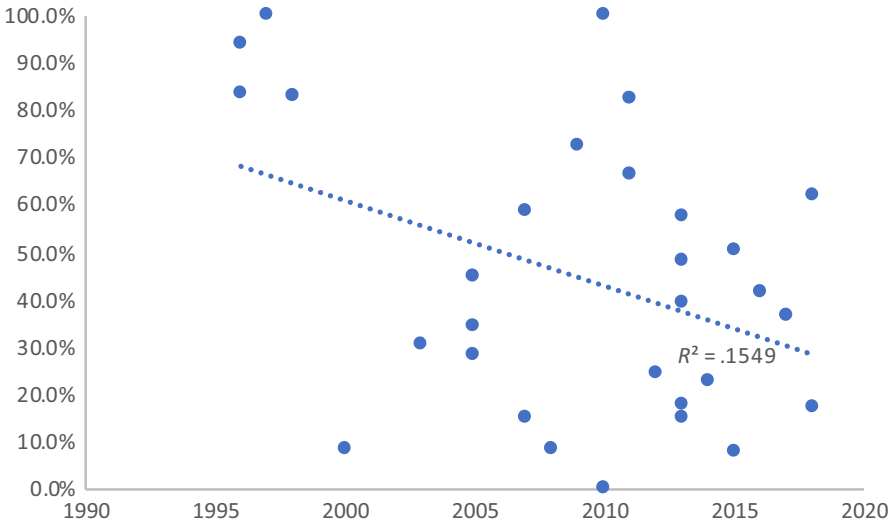

FIGURE 6 Chronological trends in use of chemical restraint as a percentage of all restrained consumers/month in year order of publication

3.5 | Methods of chemical restraint (drugs and administration)

Twelve papers reported some information on chemical restraint methods; however, this was inconsistently provided. The most consistently used benzodiazepine drugs were diazepam or lorazepam (reported in seven papers), while the most commonly reported antipsychotic drugs were droperidol or haloperidol (reported in 10 papers). From information on 3,005 consumers for whom details of chemical restraint delivery were available, the most commonly used route was intramuscular (75.2%), followed by intravenous (21.5%), then oral (3.3%).

3.6 | Adverse events

These were variably reported in nine papers, with the type of adverse event inconsistently described. Adverse events reflected three constructs (adverse health outcomes for the consumer; injury to staff; or the need for additional medication to induce sedation or sleep). Where data were available, adverse health events occurred for approximately 1%–2% consumers, and when they were described, they largely consisted of prolonged sedation or respiratory support provided until consumers gained consciousness. The prevalence of injuries to staff ranged from 2% to 10%; however, the nature of injuries was rarely described. The need for additional sedation occurred in 2%–4% consumers. As the dosage of chemical restraint was rarely described nor was its calculation with respect to consumer size, prolonged sedation cannot be inferred as over-medicating, nor can the need for additional sedation be inferred as under-medicating.

3.7 | Meta-analysis

The prevalence data from papers that reported any event of restraint, compared with the total number of admissions to facilities (all data standardized to one month), were interpreted as a meta-analysis, using a random-effects model (indicated by the significant Q statistic (degrees of freedom in brackets); $Q(31) = 2{,}728$,

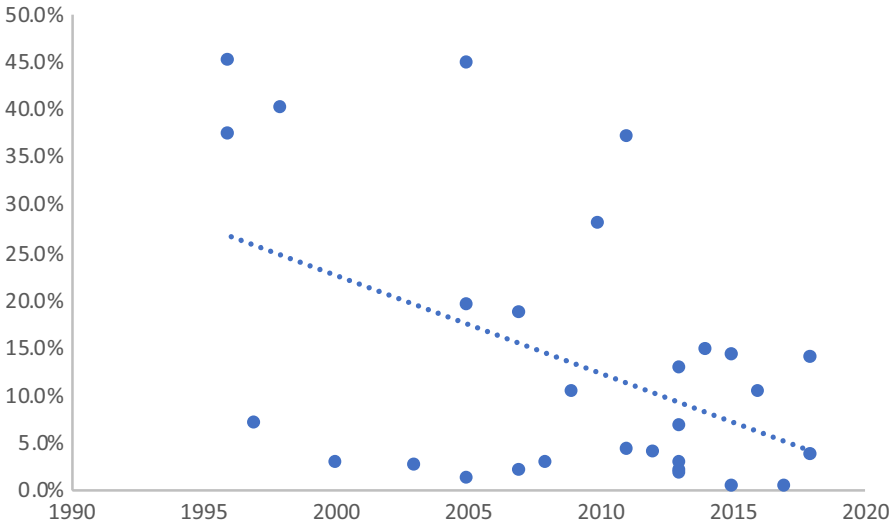

FIGURE 7 Chronological trends in use of chemical restraint as a percentage of all admitted consumers/month in year order of publication

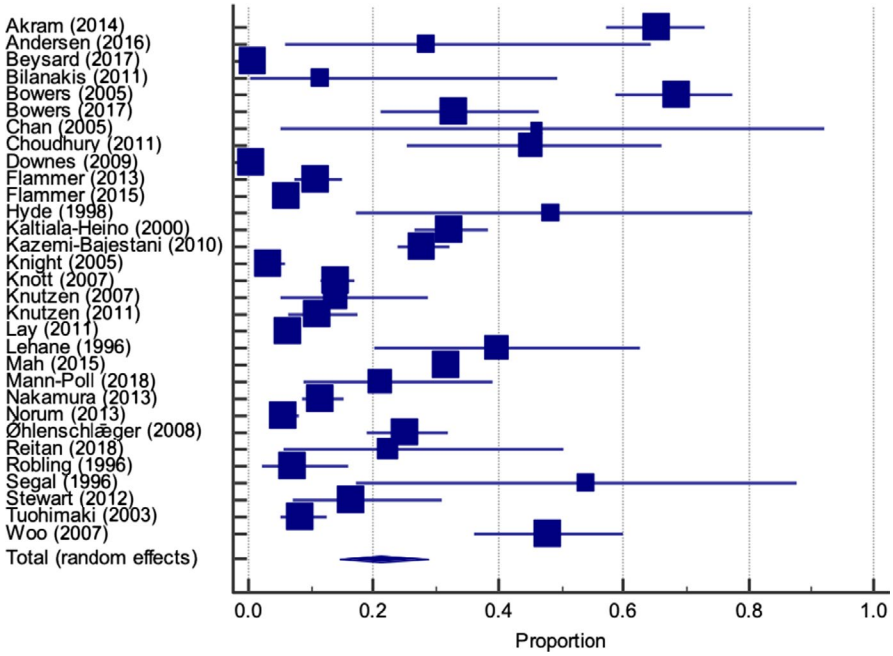

FIGURE 8 Forest plot of random-effects meta-analysis of any event of restraint as a proportion of all admissions

p = .009). The overall weighted prevalence of the use of any physical or chemical restraint was 21.6% (95% CI 14.9 to 29.1; see Figure 8 for the forest plot). The appropriateness of using the random-effects model was supported by the funnel plot, which indicated a non-random distribution of standard errors (all positively skewed), and evidenced by a high I^2 statistic for inconsistency (98.9% (95% CI 98.7 to 99.0); see Figure 9 for the funnel plot).

The data from papers that reported chemical restraint only, compared with the total number of admissions to facilities (all data standardized to 1 month), were also interpreted as a meta-analysis, again using random-effects model (Q statistic 1,625.3 (df = 30), p = .006; Raboch et al., 2010 was removed as this paper did not report this information at consumer-level). The overall weighted prevalence of use of chemical restraint by admission was 9.5% (95% CI 5.8 to 13.9; see Figure 10 for the forest plot). The appropriateness of using the random-effects model was supported by the funnel plot, which indicated a non-random distribution of standard errors (again all being positively skewed) as evidenced by a high I^2 statistic for inconsistency (98.1% [95% CI 97.8 to 98.4]; see Figure 11 for the funnel plot).

The data from papers that reported chemical restraint, compared with any restraint (all data standardized to one month), were also interpreted as a meta-analysis, again using random-effects model (Q statistic 602.5 [df = 30], p = .008; Raboch et al., 2010 was again removed). The overall weighted prevalence of use of chemical restraint was 47.8% (95% CI 35.6 to 60.1; see Figure 12 for the forest plot). The appropriateness of using the random-effects model was again supported by the funnel plot, which while it appeared to have a more normal distribution of standard errors was still asymmetrical. There was a more normal distribution of standard errors in the funnel plot, indicating slightly more positive skew than negative skew (I^2 statistic for inconsistency (95.0% [95% CI 93.8 to 95.9]; see Figure 13 for the funnel plot).

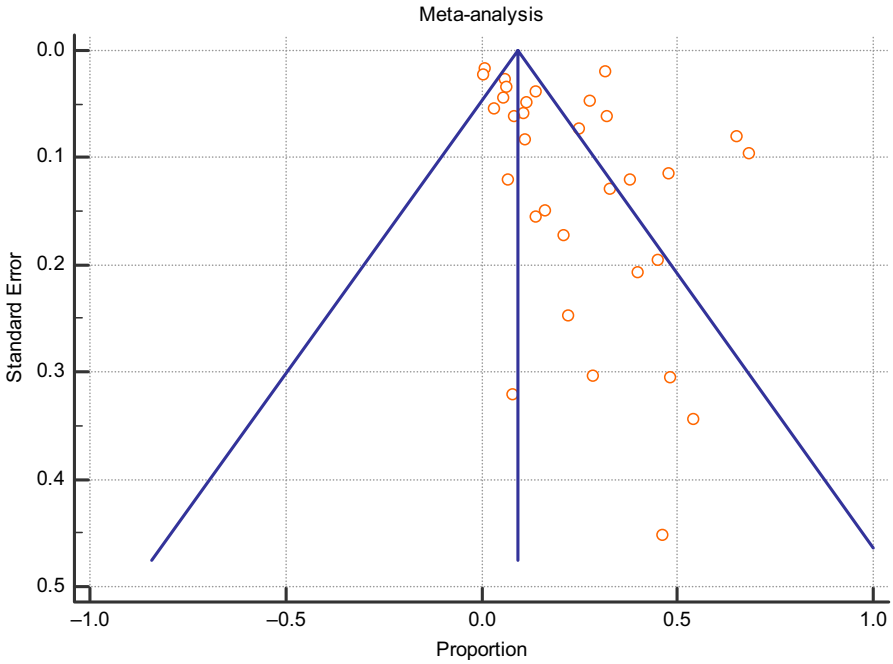

FIGURE 9 Funnel plot outlining the heterogeneity of the meta-analysis of data for any restraint compared with total admissions

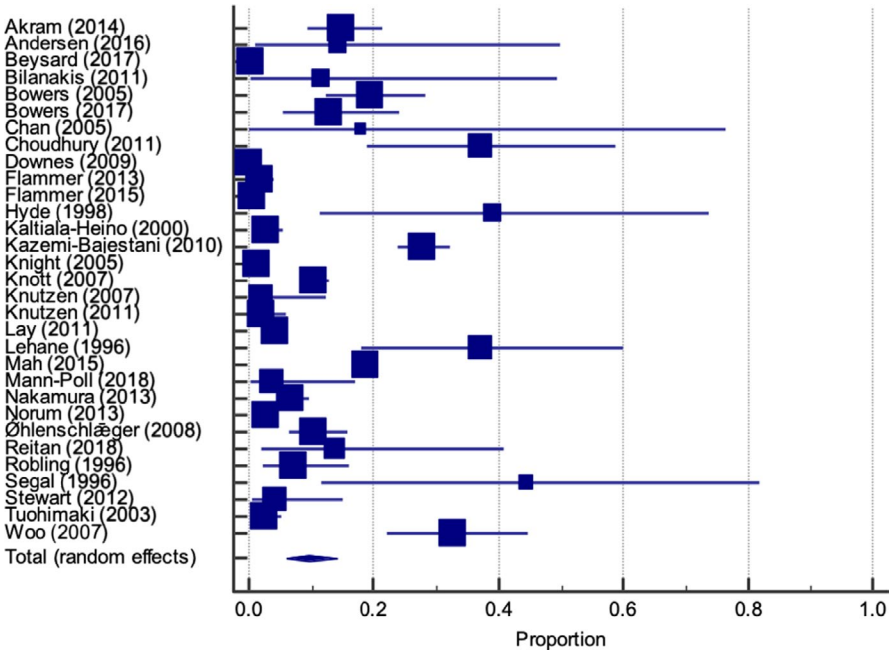

FIGURE 10 Forest plot of random-effects meta-analysis of chemical restraint as a proportion of all admissions

4 | DISCUSSION

This is the first paper to systematically review the prevalence of use of chemical restraint to manage agitation, aggression or violence in consumers in acute psychiatric or emergency settings. The median prevalence of use of any restraint was 21.2%. The median prevalence of people who were chemically restrained, of all people restrained in any manner, was 43.1%, and of all people admitted to facility(ies), the median prevalence of chemical restraint was 7.4%. The meta-analysis showed that chemical restraint was reported for 9.5% of all consumers in the data set and 47.8% of restrained consumers. The chemical restraint rate of 9.5% is within the range reported in a systematic review of restraint in psychiatry, where the prevalence of the use of restraint was 3.8%–20% (Beghi et al., 2013). Also consistent with Beghi et al.'s (2013) systematic review was the significantly larger percentage of men than women in both the overall audit sample and the sample of those experiencing chemical restraint.

This review highlighted large country-by-country variations in prevalence of use of chemical restraint (Figures 2, 3 and 4), consistent with the findings of international studies (such as the EUNOMIA Project), which have reported wide variations in chemical restraint use (Raboch et al., 2010). This potentially reflects differences in legislation, sampling frames, methods of data capture and recording, accuracy and comprehensiveness of recording, healthcare settings

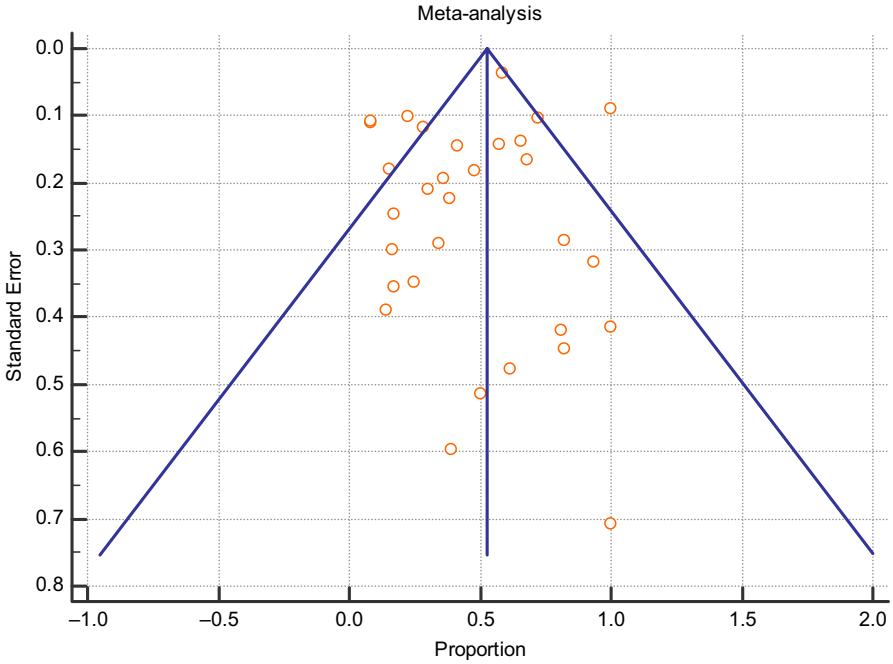

FIGURE 11 Funnel plot outlining the heterogeneity of the meta-analysis of data for chemical restraint as a proportion of any restraint (physical or chemical)

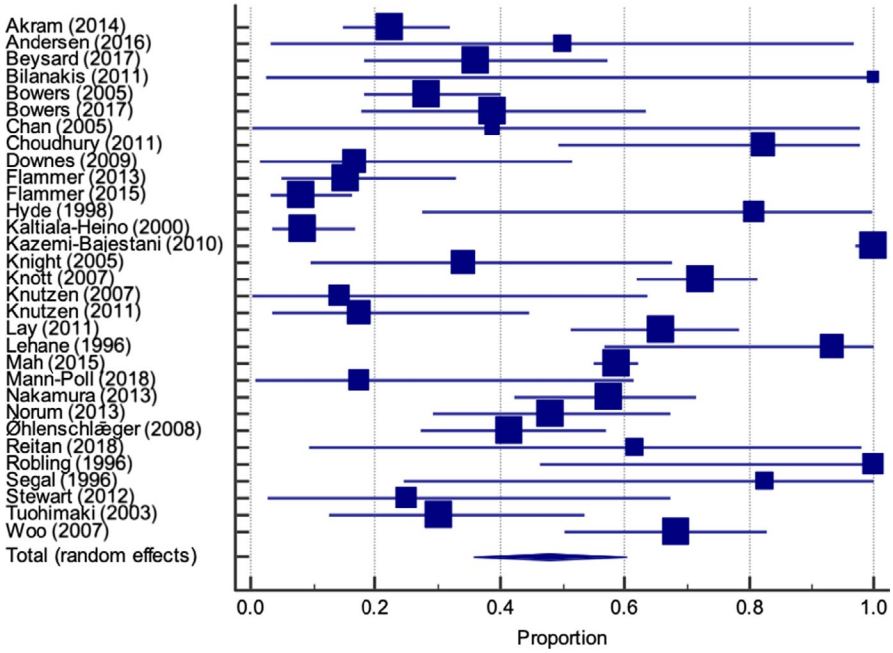

FIGURE 12 Forest plot of random-effects meta-analysis of chemical restraint as a proportion of all restrained consumers

(location, policies and practices, workload and staffing levels, staff education and attitudes to use of restraints), potential availability of psychiatric beds, time periods of audits and definitions (McSherry, 2017).

The significant decrease in prevalence of chemical restraint use over the period of the review potentially reflects changes over time in nursing practices and emergency management strategies (McSherry, 2017; Figures 5, 6 and 7). In Australia for the past decade, there has been a focus on the reduction of seclusion and restraint through the use of comprehensive de-escalation processes and the use of restraint in any form as a last resort (Gerace & Muir-Cochrane, 2019). Another initiative that may have had an impact internationally has been the introduction of the Six Core Strategies developed by the National Association of State Mental Health Program Directors (2005) in the USA. These are based around leadership, the use of data, workforce training, the use of specific reduction tools, consumer roles and debriefing. Of note was that 17 of the studies in this review stated that chemical restraint had not been the first line of management, but very few gave details of the first-line management strategies, or how these had failed. In order to offer least restrictive care for people with challenging behaviours and at their most vulnerable, it is necessary to examine and develop best practice evidence-based guidelines to reduce the need for forms of coercion in health care.

The decrease over time in rates of chemical restraint is notable given recent claims regarding rates of challenging behaviours in health

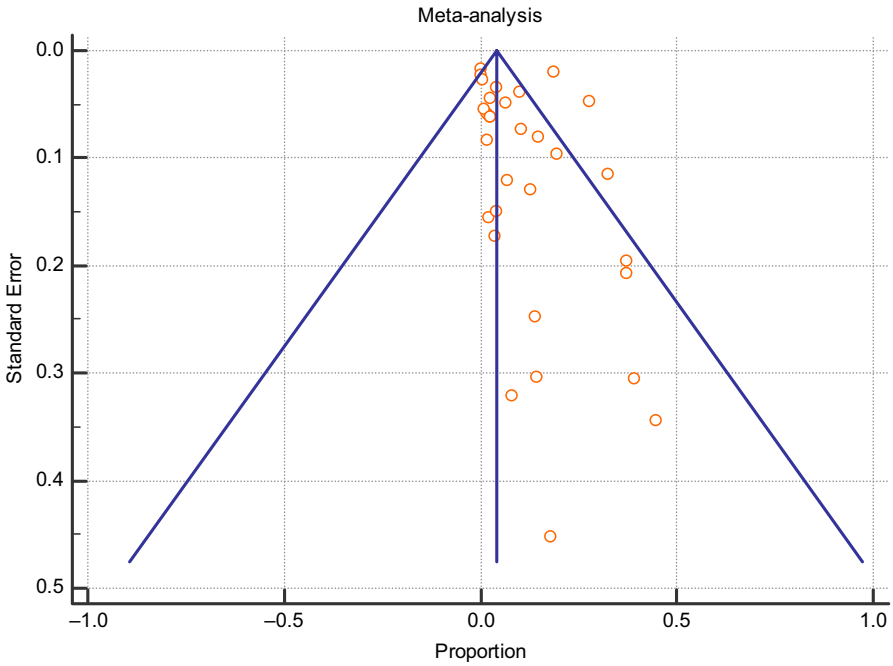

FIGURE 13 Funnel plot outlining the heterogeneity of the meta-analysis of data for chemical restraint relevant to all admissions

care. There have been recent reports of rising prevalence of violence, aggression and abuse in the healthcare sector, which threatens workers' rights to a safe and harassment-free workplace and may indicate the need for higher rates of chemical restraint (Hu et al., 2019; Osteinelli, Jajawi, Spyridi, Sayal, & Jayaram, 2018; Tangu et al., 2017; Weiland et al., 2017). These claims are potentially based on local data and highlight the need for local circumstances to be seen in an international context. Furthermore, given chemical restraint is sometimes used as an alternative to seclusion (Georgieva, Mulder, & Noorthoorn, 2013), it is unclear the extent to which seclusion may be affected by the use of chemical restraint or be used as a substitute for chemical restraint.

The review also highlights the importance of having a common denominator for comparison. To explore the data available in the included studies, we extracted three numerators and two denominators from included studies, which reported consumer-level data. This provided three quite different stories: any physical or chemical restraint applied to presenting consumers with mental health conditions; chemical restraint as part of any form of restraint to presenting consumers with mental health conditions; and chemical restraint applied to presenting consumers with mental health conditions. However differently the data were reported, the large variability was observed across countries.

The review highlighted gaps in knowledge and opportunities to improve data collection, data extraction and reporting of audits. While over 60% of studies scored well on critical appraisal of study methods (Munn et al., 2015), missing information on the delivery of chemical restraint (what was administered, how it was administered, what were the triggers for its administration), descriptors of the people to whom it was administered, the preceding management approaches (de-escalation, etc.), and information on adverse events constrained a more comprehensive review of the last 22 years of chemical restraint practices around the world. This information is important to understand if we want to introduce measures to reduce/eliminate the use of chemical restraint to fully understand the dynamics and context of the use of involuntary medication to cease challenging behaviours.

4.1 | Limitations

The review included only retrospective audit data, and data were extracted only from papers that reported a time frame of data collection and at least one denominator allowing calculation of prevalence. The included papers often reported similar rationale for conducting audits, generally reflecting lack of knowledge about local chemical and other restraint practices, concerns about adverse effects of medication such as oversedation or hypoxia, evaluation of potential biases in practices (e.g., gender, age, race), or for comparison within or between institutions. Prevalence data are also reported in cross-sectional or prospective studies in our larger data set. However, these studies were not included as they would have had different purposes, time frames and sample selection approaches, and thus would potentially have incurred different types of biases (Merlin et al., 2009). A further limitation is the time frame of study (1996–2018) given it is likely that there were changes to practice around the world, including the extent of use and nature of chemical restraint, over this length of time. Changes may also have occurred differently in different countries, although this information was not generally reported in the included papers.

5 | CONCLUSION

Our review suggests that chemical restraint is used internationally for the emergency management of approximately 1:10 adults whose behaviours cannot be managed by less restrictive means. It is not clear from the literature what the prevalence of threatened or actual

violence is in healthcare settings. Our review highlights the need for more standardized data collection, so that easier comparisons of healthcare practices can be made over time, within and between institutions, and between countries.

ACKNOWLEDGMENTS

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CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

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